

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

2 Complexity Theory (ad260)

Let $f : \Sigma^* \rightarrow \Sigma^*$ be a function on Σ -strings for some finite alphabet Σ . Say that f is a *pseudo one-way function* if it satisfies the following three conditions:

- There is a constant k such that for every $x \in \Sigma^+$, $|x|^{1/k} \leq |f(x)| \leq |x|^k$. (Here $|x|$ denotes the length of a string x).
- f is computable by a polynomial-time algorithm.
- There is no function g , computable in polynomial time, such that $f(g(y)) = y$ for all strings y in the range (i.e. image) of f .

For a pseudo one-way function f , let $L_f \subseteq \Sigma^* \times \Sigma^*$ be the following set

$$L_f = \{(x, y) \mid \exists z (z \leq_{\text{lex}} x \text{ and } f(z) = y)\}.$$

Here $\leq_{\text{lex}}$ denotes the lexicographic order on strings.

- (a) How would you modify the definition of a pseudo one-way function to obtain the definition of a *one-way function* in the sense defined by Papadimitriou?

[3 marks]

Answer: Add the additional condition that the function is injective.

- (b) Show that for any pseudo one-way function f , the language L_f is in NP.

[4 marks]

Answer: The language L_f is in NP because a string z satisfying $z \leq_{\text{lex}} x$ and $f(z) = y$ serves as a certificate of membership. The length of such a certificate is at most the length of x , since it is required to be lexicographically smaller than x . And to verify the validity of the certificate we need to check that $z \leq_{\text{lex}} x$, which is easily done and that $f(z) = y$ which can be done in polynomial time by the second point in the definition of a pseudo one-way function.

- (c) Show that for any pseudo one-way function f , the language L_f is not in P.

[4 marks]

Answer: We argue that if L_f is in P, then there is a polynomial-time computable function g satisfying $f(g(y)) = y$ for all y in the range of f , contradicting the requirement that f is a pseudo one-way function. Such a g can be computed by an algorithm that, given y , carries out a binary search among all strings of length at most $|y|^k$ looking for a string x such that $f(x) = y$.

In the following, ϕ denotes an arbitrary Boolean formula and T a list assigning a Boolean value to each variable appearing in ϕ . Fix Σ to be a suitable alphabet in

which we can write ϕ and T as well as the string “no” and consider the following function defined on all Σ -strings.

$$s(x) = \begin{cases} \phi & \text{if } x = (\phi, T) \text{ and } T \text{ satisfies } \phi \\ \text{“no”}x & \text{otherwise} \end{cases}$$

- (d) Prove that if $P \neq NP$, then s is a pseudo one-way function. [9 marks]

Answer: First of all, clearly s is computable in polynomial time. We just have to parse the input and check that it consists of a Boolean formula ϕ and an assignment T of values to all its variables—this can be done in linear time. Then, we replace all variables in ϕ by their assigned values in T and evaluate the resulting formula. By the algorithm discussed in lectures, this can be done in quadratic time.

Secondly, s satisfies the first requirement of a pseudo one-way function. To see this, note that T can be coded with one bit for each variable of ϕ , so the length of T is less than the length of ϕ . Hence, in particular, if $x = (\phi, T)$ and $s(x) = \phi$, we have $|x|/2 < |s(x)| < |x|$. On the other hand, if $s(x) = \text{“no”}x$, then $|s(x)| = |x| + 2$. In all cases, we have $|x|^{1/2} < |s(x)| < |x|^2$, for instance.

For the third requirement, we argue that if there is a polynomial-time computable function g such that $s(g(y)) = y$, then we can construct a polynomial time algorithm for deciding if a Boolean formula is satisfiable. Since SAT is NP-complete, this implies that $P=NP$, contradicting the assumption. The algorithm for SAT simply takes a Boolean formula ϕ and computes $g(\phi)$. If the result is not of the form (ϕ, T) , then ϕ is not in the range of s , so ϕ is not satisfiable. If $g(\phi)$ is of the form (ϕ, T) , we check whether T does indeed satisfy ϕ . If it does ϕ is satisfiable, otherwise it is not.
